

OBJECTIVE

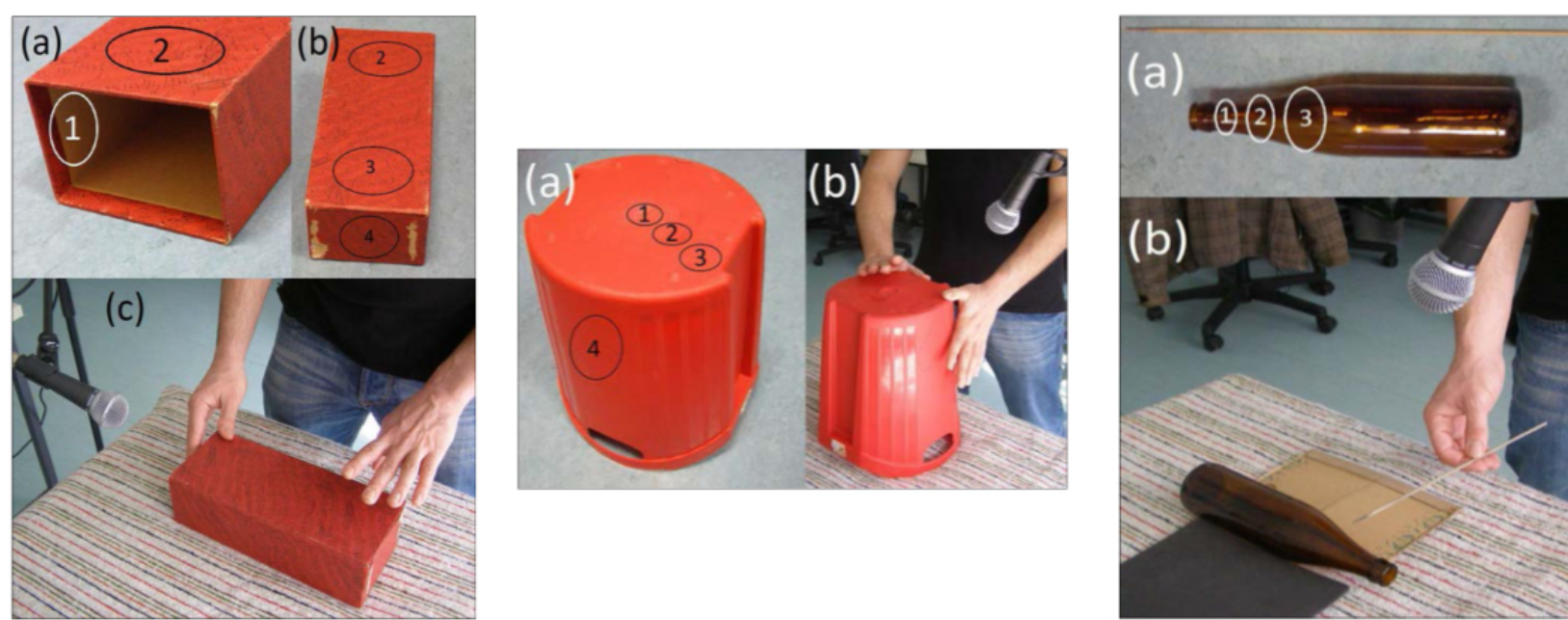
- Classification of impact sounds from the perspective of a gesture recognition system
 - Process a monophonic acoustic signal acquired by the natural stimulation of simple daily objects, in order to trigger a sound synthesis engine to produce a synthetic sound
 - Discriminating not only between different objects, but also between distinct impact locations for the same object
 - Impact sounds classification is examined in light of a sparse representations framework: relying on sparse coding and discriminative dictionary sparse coding techniques

CONTRIBUTIONS

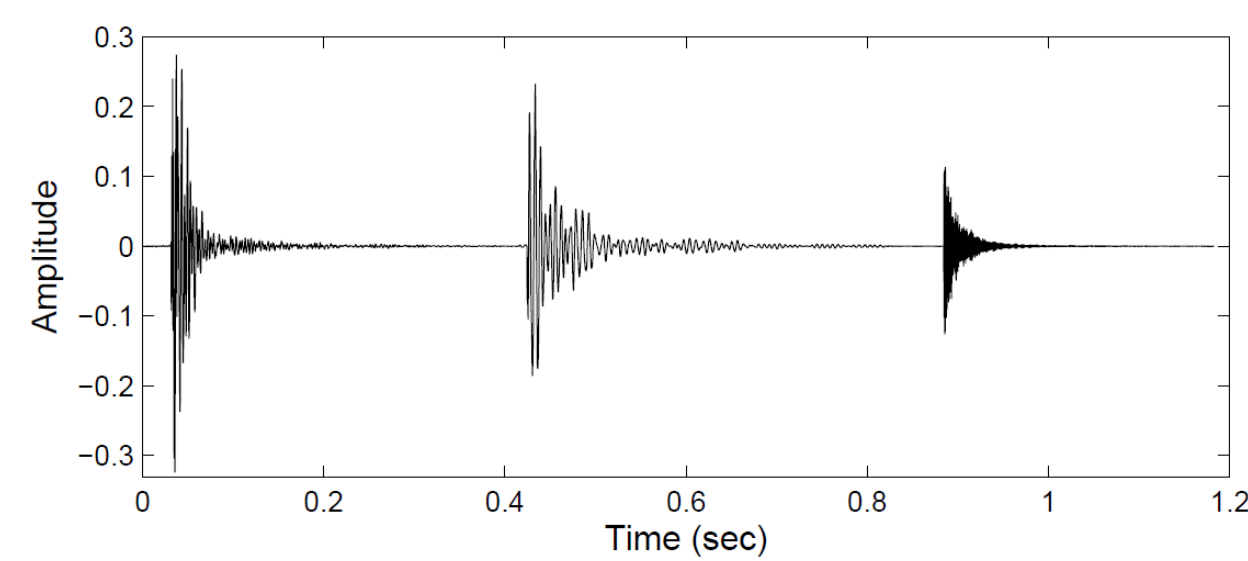
- Adopt a sparse representation framework based on learned dictionaries
- Feature extraction process is based solely on spectral magnitude information of the impact sounds: phase information is not included due to sensitivity issues in detecting the onset parameters
- Use a limited amount of training/testing data: enhance the real-time capability of the system

FEATURE EXTRACTION (1/2)

- Impact regions and microphone's placement used for recording with respect to a box made of recycled paper (left plot), a bucket made of plastic (center plot) and an empty bottle of beer (right plot)



- Each training or testing audio sample is automatically derived by applying a spectral energy based onset detection algorithm [1]
- From left to right, one instance of a strike on the box, the bucket and the bottle

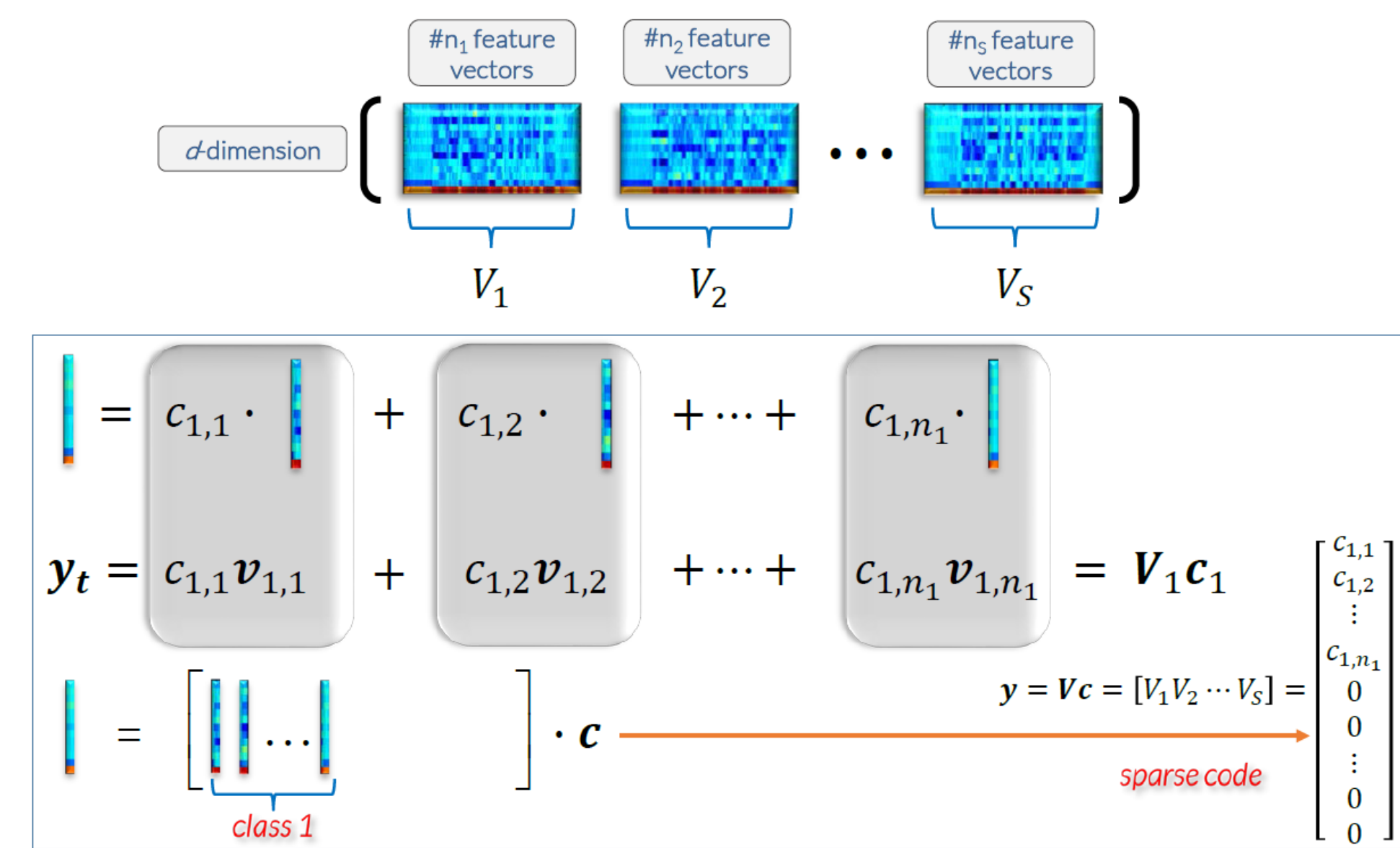


- Each impact sound is transformed via Fourier transform in the frequency domain $x_{i,j}(k)$ which corresponds to the j^{th} discrete time-domain signal of the i^{th} class
 - $k = 0, \dots, N-1$: frequency index
 - N : FFT size

FEATURE EXTRACTION (2/2)

- Capture impact sound's most dominant acoustic modes by focusing on the frequency range $f_{\min} \leq \frac{kFs}{N} \leq f_{\max}$
 - F_s : sampling frequency
- $\mathbf{x}_{i,j} = [x_{i,j}(k_{\min}), \dots, x_{i,j}(k_{\max})]^T \in \mathbb{R}^d$: vector associated with an impact sound, corresponding to the magnitude information of the computed FFT
- $\mathbf{v}_{i,j} := \mathbf{x}_{i,j} / \|\mathbf{x}_{i,j}\|_2$: normalized spectral feature vector
- $\mathbf{V}_i = [\mathbf{v}_{i,1} | \mathbf{v}_{i,2} | \dots | \mathbf{v}_{i,n_i}] \in \mathbb{R}^{d \times n_i}$: dictionary containing all the (training) feature vectors belonging to the i^{th} class
- $\mathbf{V} = [\mathbf{v}_{1,1} | \dots | \mathbf{v}_{1,n_1} | \mathbf{v}_{2,1} | \dots | \mathbf{v}_{2,n_2} | \dots | \mathbf{v}_{S,1} | \dots | \mathbf{v}_{S,n_S}] = [\mathbf{V}_1 | \mathbf{V}_2 | \dots | \mathbf{V}_S] \in \mathbb{R}^{d \times N_{tr}}$: concatenation of all the training data matrices $\mathbf{V}_i$, for $i = 1, \dots, S$
 - S : total number of classes
 - $N_{tr} = n_1 + \dots + n_S$: total number of training data

SPARSITY BASED CLASSIFICATION



Approximate sparse solution
 $\hat{c} = \arg \min_c \|c\|_1 \text{ s.t. } y_t = Vc$

OMP
 $\hat{c} = \arg \min_c \|y_t - Vc\|_2^2 \text{ s.t. } \|c\|_0 \leq K$

LASSO
 $\hat{c} = \arg \min_c \frac{1}{2} \|y_t - Vc\|_2^2 + \lambda_1 \|c\|_1 + \frac{\lambda_2}{2} \|c\|_2^2$

- Orthogonal matching pursuit (OMP) [2]: greedy sparse approximation method (low computational complexity)
 - K : number of non-zero elements in $\hat{c}$
- Least absolute shrinkage and selection operator (LASSO): enhances stability and retains correlated features via the ℓ_2 norm penalty, whilst producing a sparse feature space through the ℓ_1 norm penalty (can be solved using elastic net regression [3])
 - λ_1, λ_2 : regularization parameters

- Estimated feature vector $\hat{y}_i = V\delta_i(\hat{c})$
 - assigned to the class giving the minimum residual

$$i^* = \arg \min_i \|y_t - V\delta_i(\hat{c})\|_2^2, i = 1, \dots, S$$

- $\delta_i: \mathbb{R}^{N_{tr}} \rightarrow \mathbb{R}^{N_{tr}}$ denotes an auxiliary indicator function for each class i , where the entries of the vector $\delta_i(\hat{c})$ are all zero except for those corresponding to the i^{th} class

DISCRIMINATIVE DICTIONARY SC

- Introduce the concept of discriminative dictionary learning for impacts sounds classification based on the Fisher criterion [4]
- Enhance the discriminative capability of the estimated sparse codes
 - Enforce the sparse codes having small within-class scatter but big between-class scatter
 - Enforce each class-specific subdictionary having good reconstruction capability to the training samples from that class but poor reconstruction capability to other classes

$$\hat{\mathbf{D}}, \hat{\mathbf{C}} = \arg \min_{\mathbf{D}, \mathbf{C}} \mathcal{T}(\mathbf{V}, \mathbf{D}, \mathbf{C}) + \beta_1 \|\mathbf{C}\|_1 + \beta_2 \mathcal{R}(\mathbf{C}),$$

s.t. $\|\mathbf{d}_j\|_2 = 1, j = 1, \dots, N_{tr}$

discriminative accuracy term sparsity penalty sparse codes discrimination term

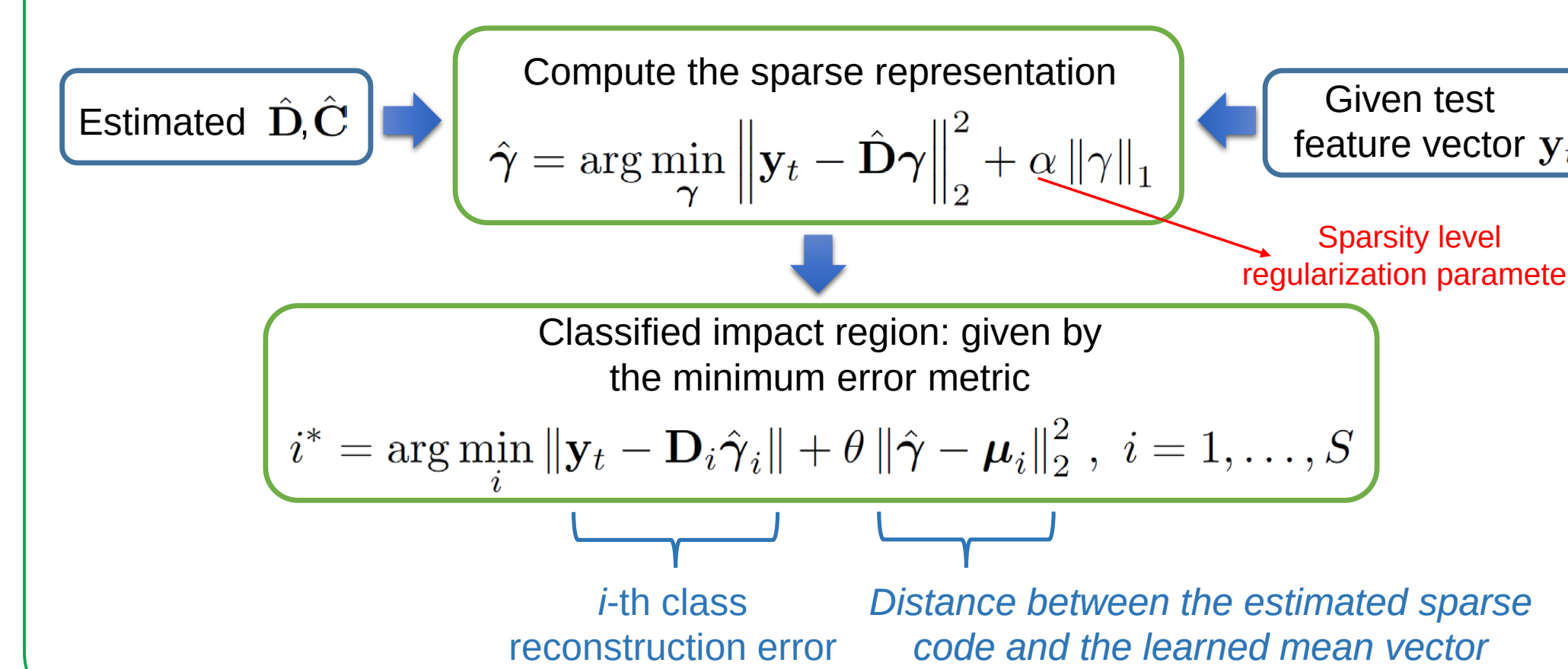
- β_1 : regularization parameter controlling the sparsity level of the corresponding sparse codes
- β_2 : regularization parameter controlling the discriminative ability imposed on the sparse codes

- The discriminative accuracy term is defined by

$$\mathcal{T}(\mathbf{V}_i, \mathbf{D}, \mathbf{C}_i) = \|\mathbf{V}_i - \mathbf{D}\mathbf{C}_i\|_F^2 + \|\mathbf{V}_i - \mathbf{D}_i\mathbf{C}_i^i\|_F^2 + \sum_{j=1, j \neq i}^S \|\mathbf{D}_j\mathbf{C}_i^j\|_F^2$$

- $\mathbf{V}_i = \mathbf{D}\mathbf{C}_i = \mathbf{D}_1\mathbf{C}_i^1 + \dots + \mathbf{D}_i\mathbf{C}_i^i + \dots + \mathbf{D}_S\mathbf{C}_i^S$
 - $\mathbf{V}_i \in \mathbb{R}^{d \times n_i}, \mathbf{D}_i \in \mathbb{R}^{d \times Z_i}, \mathbf{C}_i^j \in \mathbb{R}^{Z_j \times n_i}$ and the size of $\mathbf{D}$ is $Z = Z_1 + \dots + Z_j + \dots + Z_S$
 - If $\mathbf{D}_i$ corresponds to the class i , we expect that $\mathbf{V}_i$ will be correctly represented by $\mathbf{D}_i \Rightarrow \mathbf{D}_i$ will have not only the minimal but also very small representation residual for $\mathbf{V}_i$, while other sub-dictionaries will have big representation errors of $\mathbf{V}_i$

- Fisher discriminant criterion $\Rightarrow$ minimize the within-class scatter $S_w(\mathbf{C}) = \sum_{i=1}^S \sum_{c_j \in \mathbf{C}_i} (\mathbf{c}_j - \mu_i)(\mathbf{c}_j - \mu_i)^T$ between the sparse codes $\mathbf{C}$, whilst maximizing the between-class scatter $S_b(\mathbf{C}) = \sum_{i=1}^S n_i(\mu_i - \mu)(\mu_i - \mu)^T$
 - μ_i : mean vector of $\mathbf{C}_i$
 - μ : mean vector of $\mathbf{C}$
- Fisher discriminant criterion $\Rightarrow \mathcal{R}(\mathbf{C}) = \text{tr}(S_w(\mathbf{C})) - \text{tr}(S_b(\mathbf{C})) + \tau \|\mathbf{C}\|_F^2$
 - τ : parameter associated with the stability term $\|\mathbf{C}\|_F^2$



EXPERIMENTAL RESULTS

- All the recordings took place in a room of dimensions $8 \times 7 \times 2.5 \text{ m} \Rightarrow$ cardioid dynamic microphone (Shure SM 58) plugged into an external USB sound card
- The total number of classes is eleven: each object's impact region defines a class
- Sampling frequency 44100 Hz $\Rightarrow$ downsampled at 22050 Hz before the spectral processing
- 64-bit MATLAB R2015a programming environment: quad-core Intel Core i7-2670 with 8GB RAM
- Length of the spectral analysis window: 20, 9, 6 and 4 msec (FFT size equals the length of the analysis window)
- Use the most informative frequency content ranging from 0 to 5000 Hz
- Following a cross-validation procedure for all the models' parameters: $K = 5, \lambda_1 = 10^{-3}, \lambda_2 = 10^{-5}, \beta_1 = 0.05, \beta_2 = 10^{-3}, \alpha = 0.01, \theta = 10^{-3}$
- Compare nearest neighbour (NN) vs. sparse coding using OMP (SRC-OMP) and LASSO (SRC-LASSO) vs. discriminative dictionary sparse coding (DDSC)

TABLE I
MEAN CORRECT CLASSIFICATION RATES OF NN VS. SRC-OMP VS. SRC-LASSO VS. DDSC AVERAGED OVER ALL CLASSES. THE LENGTH OF THE PROCESSING WINDOW IS 20 MSEC.

# train. signals per class	NN	SRC OMP	SRC LASSO	DDSC
all	94.66	95.76	95.69	95.94
20	93.93	95.02	94.98	95.00
15	94.37	95.53	95.26	96.42
10	93.07	93.88	93.30	94.47
5	91.21	92.24	92.13	92.30
2	86.98	87.92	88.54	88.04

- Correct impact sounds classification rates as a function of the class index using an 20 msec processing window. The amount of training data equals (i) all the available training signals, (ii) 20, (iii) 15, (iv) 10, (v) 5, and (vi) 2 signals, respectively

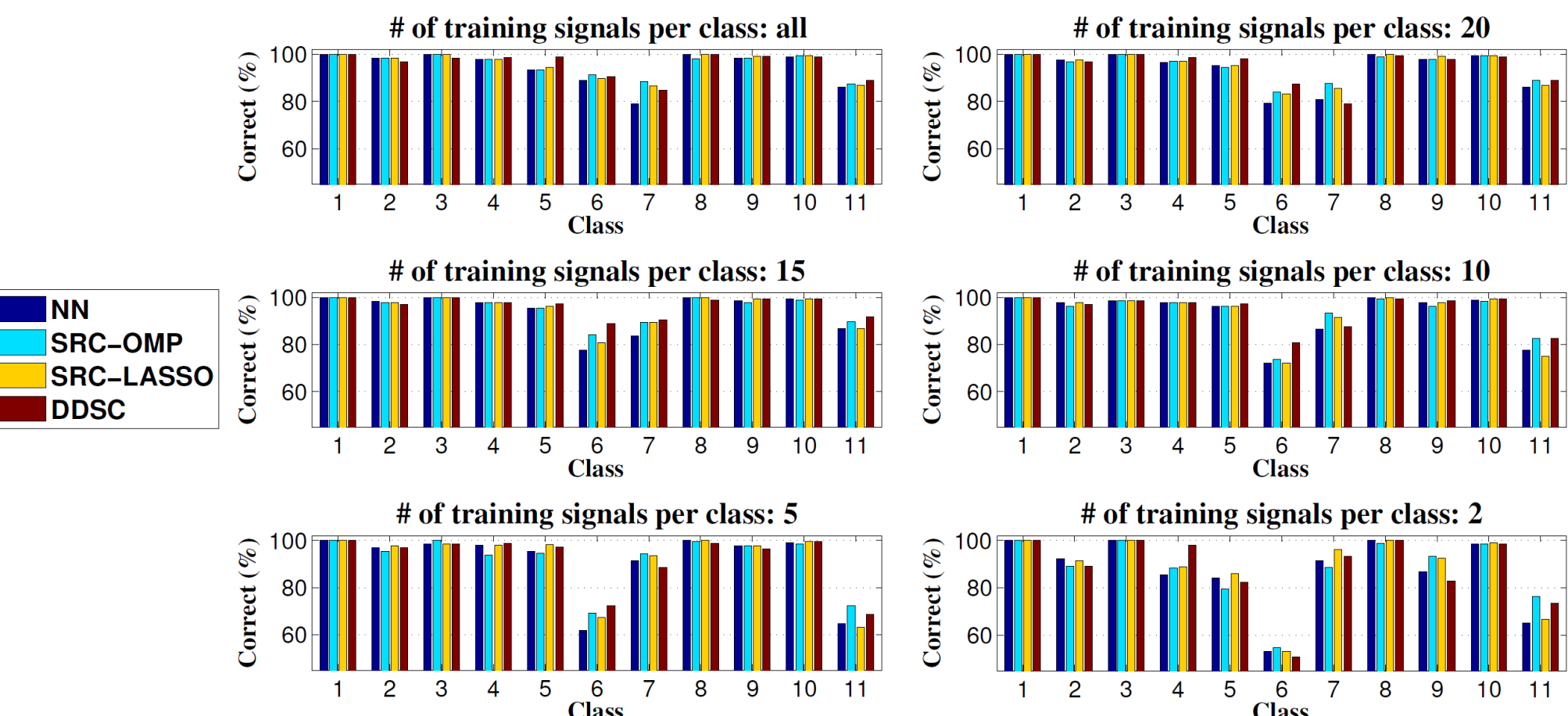


TABLE II
SRC-OMP VS. SRC-LASSO COMPUTATION TIME IN MSEC.

proces. win. (msec)	SRC-OMP	SRC-LASSO
20	0.3699	0.5163
9	0.3267	0.4543
6	0.3092	0.4273
4	0.3021	0.4185

REFERENCES

- [1] N. Stefanakis, et al., "Instantaneous detection and classification of impact sound: turning simple objects into powerful musical control interfaces," in *11th Sound and Music Computing Conference*, Athens, Greece, September, 2014.
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- [3] H. Zou and T. Hastie, "Regularization and variable selection via the elastic net," *Journal of the Royal Statistical Society, Series B*, vol. 67, pp. 301-320, 2005.
- [4] M. Yang, et al., "Sparse representation based Fisher discrimination dictionary learning for image classification," *Int. Journal of Computer Vision*, vol. 109, pp. 209-232, September 2014.